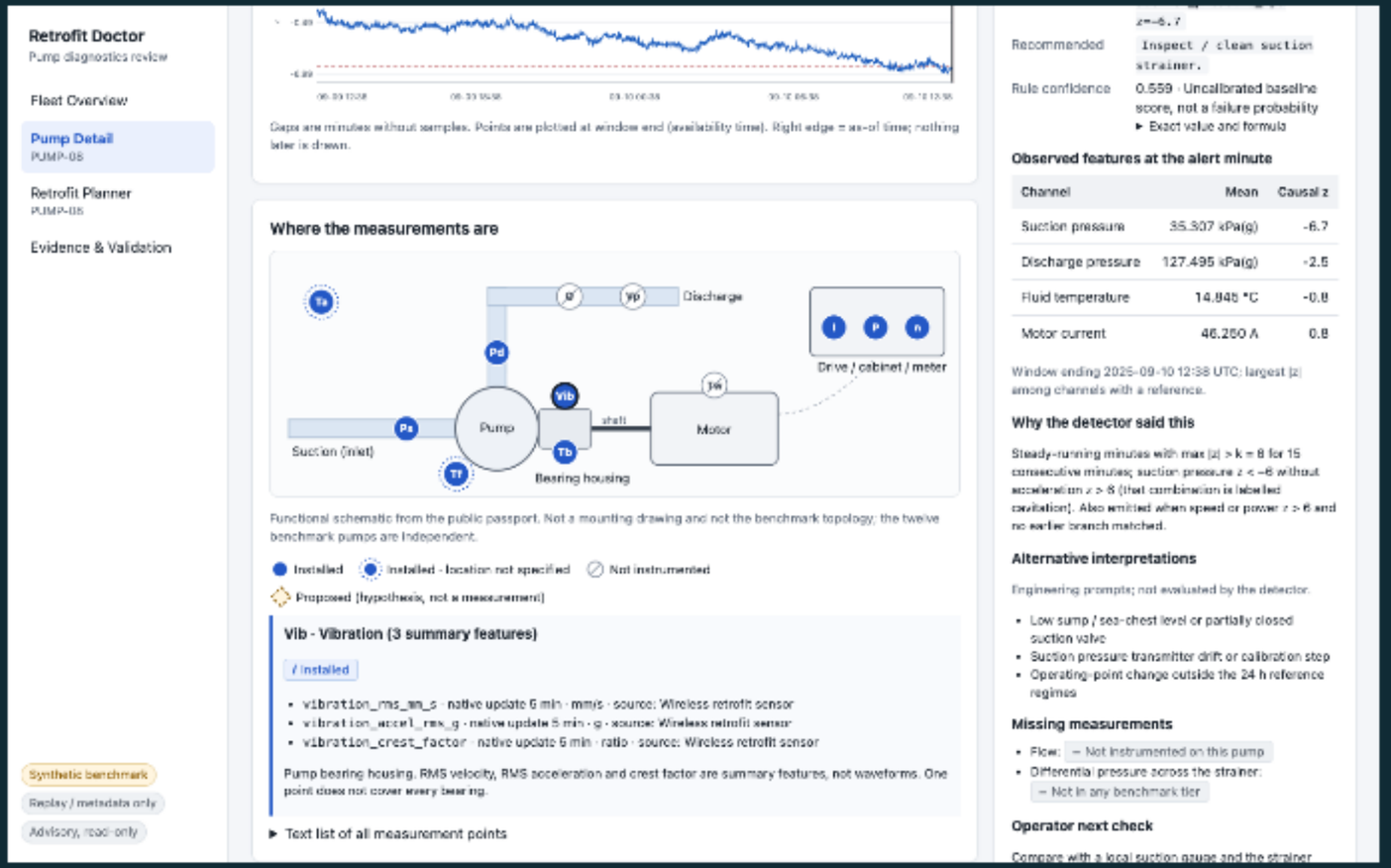
Retrofit Doctor

Pump diagnostics and retrofit review

Synthetic benchmark demonstration
Since AI 2026



Twelve independent pump assets

Split	Assets	Demo access
Training	PUMP-01–07	Development and selection
Validation	PUMP-08–09	Labeled replay and evaluation
Test	PUMP-10–12	Equipment metadata only

Different sensor coverage and observation weeks.
Site grouping does not imply hydraulic coupling.

Diagnostic evidence and limits

Validation example	Causal B1 output	Retrospective result
PUMP-08-E01	Suction strainer clogging	Correct fault type
PUMP-08-E02	Unknown cause	Event matched; unresolved
PUMP-08-E04	No matched alert	Pressure-sensor noise missed

Detection does not establish the correct cause.

Synthetic validation. Evaluation labels stay separate from live-clock evidence.

Causal baseline validation

Metric	Causal B1	Original B1
Composite score	76.01	78.01
Events detected	8 / 9	8 / 9
Correct fault type	4 / 9	4 / 9
False alert episodes	1	1

Original B1 used future reference data in early predictions.

Nine new diagnostic candidates rejected; causal B1 retained.
Reproducibility requires the pinned environment and seed 0.

Synthetic validation, 9 scored events. No field validation.

Measured value of sensor information

Same causal B1 detector · 3 train assets · 10 original events

Information available	Detected / 10	Correct type / 10	False + excess
All installed groups	9	2	6
Without vibration / bearing temperature	8	0	6
Without suction pressure	9	2	6
Without flow / power	8	3	5

Restoring each group reproduces the original result.

Vibration / temperature helps overall; other effects are mixed.
More sensors do not guarantee better diagnosis.

Synthetic information ablation. No physical installation benefit or ROI established.

Proposed measurement package

Example: a missing bearing-housing measurement point

Element	Proposed implementation	Site decision needed
Measurement	Vibration / temperature on pump bearing housing	Location, rigid mount, measurement bandwidth
Link and power	Cable to local IO, or radio after survey	Routing / power, or battery service and coverage
Gateway	Local enclosure, buffer and advisory analysis	IO, power, environment, maintenance owner
Engineering display	Existing data + new point with quality and time	Tag mapping, freshness, review workflow

Proposed field architecture. Current demo: static synthetic replay.
Reuse existing signals where sufficient; controls remain independent.

No hardware built • No selected supplier, fixed BOM or certification claim

A retrofit must justify its full cost

PUMP-08 has no flow channel. Flow assessment is a hypothesis:
pipe suitability, reference checks and diagnostic value remain unverified.

Illustrative redundant-pump case	EUR
Initial incremental installation cost	4,500
Annual benefit / nuisance checks / upkeep	300 / 1,800 / 700
Annual net effect	−2,200
Five-year incremental value (0% discount)	−15,500

Defer permanent retrofit under these assumptions.

Hypothetical standby case; not this pump’s topology, a quote or measured savings.

Pilot pathway

01 Observe existing data in shadow mode

02 Check whether warnings support useful action

03 Trial the smallest measurement that resolves a gap

04 Verify installation, integration and maintenance

05 Decide using evidence, full cost and nuisance burden

Proposed pilot. Benefits, physical feasibility and applicable approvals remain unvalidated.